ASSESSMENT REMIT

Name			
ID#			
Semester	Fall 2026	**Section**	
School	School of Engineering, Architecture and Interior Design (EAID)		
Program	Bachelor of Science in Information Technology (BS IT)		
Major	Information Technology		
Course Code	**ITEC 300**		
Course Title	Computer Communications and Networking		
Level	☐ Remedial ☐ Bridging √ Undergraduate ☐ Graduate		
Assessment Title	Assignment — University Network Design and Evaluation		
Weighting	20 Marks		
Instructor	Dr. Maher Alrahhal		
Hand Out Date	26/10/2026		
Due Date & Time	Date:30/11/2026	Time: 10:00 – 11:50 AM	
Feedback Post Date	07/12/2026		
Assessment Mode	☐ Individual √Group ☐ Closed Book √ Open Book		
Assignment Length *(for assignment-based assessments)*	5–8 pages, excluding cover and references		
Submission Mode	☐ Online by Email √ Online through LMS | TurnItIn ☐ Hard Copy		
Document Type	☐ Words √ PDF ☐ Artwork ☐ Software ☐ Other		

GENERAL REQUIREMENTS (GROUP ASSIGNMENT)
Students must work in groups of 3–4 members.
Each group must complete all assignment questions (Q1–Q7).
The assignment must be based on the provided university networking scenario.
Students must submit a report of 5–8 pages, excluding the cover page and references.
The report must include clear explanations, appropriate network diagrams, and justifications for the proposed solutions.
All group members must contribute to the assignment and clearly identify their individual responsibilities.
Each student must participate in a short individual oral discussion to demonstrate their understanding of the network design.
Written assignment marks will be awarded to the group, while oral discussion marks will be awarded individually.
The report must be submitted as a PDF through LMS / Turnitin before the specified deadline.
All external information, diagrams, and references must be acknowledged using APA referencing style.
The use of generative AI to write the assignment is not permitted unless explicitly authorized by the instructor.

All submissions must comply with Amity University Dubai's Academic Integrity and Academic Misconduct Policy.

ASSESSMENT:

Assignment Scenario

A university plans to upgrade its network infrastructure to provide reliable wired and wireless connectivity.

The university has the following requirements:

Location	Network Requirements
Computer Laboratory	30 desktop computers
Library	20 desktop computers and 40 laptops
Administration Office	15 desktop computers and 2 network printers
Classrooms	Wireless connectivity for 60 laptops
University Buildings	Two buildings located 500 meters apart
University Server	One central server providing shared files and network services

The university requires a network that is reliable, secure, easy to manage, and suitable for future expansion.

Task: Work as a team of network designers to propose, explain, and evaluate a suitable university network infrastructure.

Part A – Network Infrastructure and Protocols (10 Marks)

Question 1 – Identify Network Devices (3 Marks)

Identify five networking **devices or components** required for the university **network infrastructure**.

For each device or component:

a) State its name.

b) **Explain** its main function.

c) **Identify** where it will be used in the university network.

Complete the following table:

Device / Component	Main Function	Location / Purpose
1.		
2.		
3.		
4.		
5.		

Question 2 – Network Services and Protocols (4 Marks)

Apply principles of computing to **identify and explain** the network **services and protocols** required to support students and staff.

Consider the following four services and protocols:

1. DHCP
2. DNS
3. TCP/IP
4. HTTPS

For each service or protocol:

a) **Explain** its main function.

b) Provide one practical example of how it will be used in the university **network infrastructure**.

Example: DHCP can automatically assign IP addresses to student laptops when they connect to Wi-Fi.

Question 3 – Network Communication Mechanisms (3 Marks)

A student connects a laptop to the university Wi-Fi and opens the university website.

Apply principles of computing to **explain** the basic network communication **mechanisms, services, and protocols** involved.

Answer the following:

a) How does the laptop connect to the university network?

b) How does the laptop obtain an IP address?

c) How does the laptop find and access the university website?

Your answer should **identify** the relevant networking devices, services, and protocols.

Part B – Network Design and Evaluation (10 Marks)

Question 4 – Design the University Network (4 Marks)

Work in a team to **design a network** that meets the requirements of the **given university problem**.

Your network design must include:

- Desktop computers and laptops.
- Network switches and wireless access points.
- A router or gateway.
- A central server.
- Appropriate transmission media.
- A connection between the two university buildings.

Required:

a) **Design** and draw a clear network topology diagram using Cisco Packet Tracer, draw.io, or another suitable tool. **(2 Marks)**

b) **Explain and justify** why your selected networking **devices** and transmission media are suitable for the university network. **(2 Marks)**

Your diagram must clearly show how the main network components are connected. A working network simulation is not required.

Question 5 – Evaluate the Proposed Network (2 Marks)

Evaluate the proposed network **design**, including the suitability of its **devices and protocols**, based on the following criteria:

a) **Reliability (0.5 Mark):** Evaluate how your network design provides reliable communication for university users.

b) **Security (1 Mark):** Identify two security measures or protocols that can protect the university network and explain their purpose.

c) **Scalability (0.5 Mark):** Evaluate how the network design can support future expansion when more computers, classrooms, or students are added.

Question 6 – Teamwork and Individual Contributions (1 Mark)

Work in a team to complete the network **design and evaluation** for the **given university problem**.

Explain how group members collaborated, divided responsibilities, and contributed to the final network design.

Complete the following table:

Student Name / ID	Assigned Responsibility	Work Completed
Student 1		

Student 2		
Student 3		
Student 4 (if applicable)		

Each group member must contribute to the assignment and understand the final network design.

Question 7 – Individual Oral Discussion (3 Marks)

Each student must participate in a short oral discussion with the instructor to **explain and evaluate** the proposed university network design.

During the discussion, students should demonstrate their understanding of the selected **networking devices, protocols, and network design decisions**.

Assessment Criteria:

a) **Technical Understanding (1 Mark):** Correctly explain the functions of selected networking devices and protocols.

b) **Design Evaluation (1 Mark):** Explain and justify network design decisions and evaluate their suitability for the university.

c) **Individual Contribution (1 Mark):** Demonstrate understanding of assigned responsibilities and explain how individual work contributed to the team's network design.

Note: Oral discussion marks will be awarded individually based on each student's understanding and performance.

ANSWER GUIDELINES:

Students are expected to:

- Answer all seven questions (Q1–Q7).
- Work in groups of 3–4 students.
- **Apply principles of computing** to **identify** networking devices, **mechanisms, services, and protocols**.
- **Work in a team** to **design a network** using a clearly labeled network topology diagram.
- **Explain and justify** the selected devices and transmission media.
- **Evaluate** the proposed network in terms of reliability, security, and scalability.
- Clearly **explain** each group member's responsibilities and contributions.
- Participate individually in the oral discussion and demonstrate understanding of the proposed network.

Presentation Requirements

- Include the names and student IDs of all group members.
- Organize the written report according to Questions 1–6.
- Complete Question 7 through an individual oral discussion with the instructor.
- Use clear technical explanations and practical examples.
- Create the network diagram using Cisco Packet Tracer, draw.io, or another suitable tool.
- Include references using APA style.
- Submit one complete report per group in PDF format.

REFERENCING STYLE:

Students should use the American Psychological Association style (APA) as per the following guidelines: https://apastyle.apa.org/style-grammar-guidelines/references/examples

PLAGIARISM & GENERATIVE AI (FOR ASSIGNMENT BASED ASSESSMENTS):

Amity University Dubai is committed to a policy of honesty in all its operations, especially those pertaining to academic matters. It is your responsibility to use appropriate references and apply correct referencing practices. Unless explicitly stated otherwise, the use of generative AI to write an assignment is not permitted. Submission should be yours and should not be in breach of AMUD's Academic Integrity and Academic Misconduct Policy.

LATE SUBMISSIONS (FOR ASSIGNMENT BASED ASSESSMENTS):

You are advised to strictly adhere to the deadline. While Amity University Dubai might consider extenuating circumstances to support students who are experiencing unforeseen issues impacting their ability to complete assessments on time, any anticipated submission delay must be communicated in advance to the instructor who will consult with the Program Leader and the Dean before a permission is granted. All permissions must be based on valid reasons backed up by concrete evidence. For more details, you are advised to consult the latest version of AMUD's Academic Integrity and Academic Misconduct Policy.

FEEDBACK PLAN:

The preparation for tasks included in this assessment will be supported by formative feedback during the class regular sessions and related office hours. Summative written feedback will be provided on the date indicated earlier.

ASSESSED COURSE LEARNING OUTCOMES:	RELATED QUESTIONS	MAX SCORE	MARK

CLO 3: Apply principles of computing to identify the basic mechanisms, services and protocols for the organizational network infrastructure.	Q1–Q3	**10**	
CLO 4: Work in a team to evaluate the devices and protocols to design a network using suitable explanations for the given problem.	Q4–Q7	**10**	
	TOTAL	**20**	

EMBEDDED PLO DIRECT ASSESSMENT QUESTIONS? ☑ **YES** ☐ **NO**

[IF YES, COMPLETE THE TABLE BELOW]

PLO 1	**EMBEDDED QUESTIONS**
PLO1: Analyze a complex computing problem and to apply principles of computing and other relevant disciplines to identify solutions.	CLO 3: Q1–Q3 (10 marks). CLO 4: Q4–Q7 (10 marks). Total: 20 marks.

SCORING RUBRIC			**LEVEL OF ACHIEVEMENT**			
	DESCRIPTOR	**WEIGHT**	**4 [87%-100%]**	**3 [73%-86%]**	**2 [60%-72%]**	**1 [0%-59%]**
1	CLO 3: Network Devices (Q1)	3	Correctly identifies five devices, functions, and locations.	Mostly correct with minor errors.	Some correct devices; incomplete explanations.	Incorrect or missing devices and explanations.
2	CLO 3: Services and Protocols (Q2)	4	Accurately explains all four services/protocols with practical examples.	Mostly correct explanations and examples.	Partial understanding with several errors.	Incorrect or missing explanations.
3	CLO 3: Network Communication (Q3)	3	Clearly explains Wi-Fi connection, IP assignment, and website access.	Mostly correct communication process.	Incomplete process with some errors.	Incorrect or missing communication explanation.
4	CLO 4: Network Design (Q4)	4	Complete, clearly labeled topology with appropriate devices and strong justification.	Suitable design with minor omissions.	Partially complete design with weak justification.	Incorrect or incomplete design.
5	CLO 4: Network Evaluation (Q5)	2	Clearly evaluates reliability, security, and scalability.	Mostly correct evaluation with minor omissions.	Partial evaluation with limited explanation.	Incorrect or missing evaluation.
6	CLO 4: Teamwork (Q6)	1	Clear responsibilities and documented contributions from all members.	Mostly clear contributions.	Limited evidence of contributions.	Missing or unclear teamwork evidence.
7	CLO 4: Individual Oral Discussion (Q7)	3	Clearly explains devices, protocols, and design decisions; demonstrates strong individual understanding and contribution.	Mostly correct explanations with minor gaps.	Partial understanding and limited justification.	Unable to explain key devices, protocols, design decisions, or individual contribution.

Student Declaration

By signing this cover page, you acknowledge the instructions above (general requirements) and accept all responsibilities resulting from breaching them as per the related University policies. I hereby declare that it is my original work and I have duly acknowledged and referenced the source of any quotes, thoughts or ideas incorporated in the work.

No.	Student Name	Student ID	Student Signature	Date
1				
2				
3				
4				

To be filled by faculty

FEEDBACK FROM FACULTY:

Faculty Name and Signature **Date**